Project 1:Data Analysis

Professor:
Omar Mendoza Montoya

Miguel Angel Rivas Torres | A01644628
Alan Fernando Bravo Pimentel | A01645691
Pedro Manuel Montes Valle | A01741659

Scripts:

https://drive.google.com/drive/folders/1Zb0zmzpc9lJymJdOXCPnmgBKm1Uc3GX8?usp=sharing

---

Description of the implemented application

The program is intended to serve as a personal motion detector capable of telling apart the following movements from one another: 1) Rest, 2) Walking, 3)Side-to-side waist movement, 4) Jumping, 5) Squats, 6) Kicks. This classification is carried out in real time through Phyphox's acceleration sensors and with the help of a smartphone connected to the program while being inside of the right pocket of the person performing the movements. This is done through the real-time implementation of a machine learning algorithm previously trained on data containing these movements. This application can be useful in fields such as fitness tracking, elderly care monitoring or even physical therapy.
There are certainly a few other applications that perform similar or even more complete tasks, one of them is Google Fit, this application also uses on-device machine learning algorithms to classify physical activities such as walking, running, biking based on motion sensors and through Android's Activity Recognition API. Additionally, Samsung Health is also capable of employing built-in sensors and pre-trained models to automatically detect and label activities. However, most commercially available applications only offer predefined activities with limited to no customization, as opposed to the application developed in this project which incorporated movements at will.
Admittedly, in our application we only relied on the acceleration readings provided by Phyphox whereas these couple of apps (Google Fit and Samsung Health) also incorporate timestamps for synchronization, angular velocity (if applicable) and from the data collected, a series of operations are performed in order to obtain valuable information such as statistical descriptors (mean, median, max, etc) along with frequency domain features that can be extracted from the raw data making use of the FFT.

Explanation of the features extracted from the data

During the processing of the data obtained via Phyphox, 37 features were deemed relevant for later use in the training of the machine learning models. It is important to note that the vast majority of them were triplicated so as to account for the 3 axes of acceleration (x,y,z).

- Mean (x,y,z): The mean features gives the average acceleration during the window in every axis. This was important because certain activities have different results, for instance, the jumping surely shifts the mean a lot while walking does not.

- Standard deviation (x,y,z): The std feature is useful because it captures the variability (spread) of the acceleration in every axis. It likely came in handy when distinguishing between rest and kicks.

- Kurtosis (x,y,z): The kurtosis is intended to capture peaks/spikes with ease and thus, it helps with the classification of sudden movements like kicks or jumping.

- Skewness (x,y,z): The skewness captures the asymmetry in the signal distribution and it is useful for the purposes of our application because certain movements like kicks or squats may skew the data toward the direction of the force, while rest and walking have more symmetrical readings.

- RMS: The RMS was intended to capture the overall energy, that is the combined signal strength from all axes, therefore we can expect activities like jumping to have a high RMS, rest to have the lowest one, and side-to-side or walking to fall somewhere in the middle.

- Median (x,y,z): The median is another basic statistical descriptor that in this case captures baseline orientation of the body and helps distinguish between stationary and moving activities since it is the midpoint value and is less sensitive to outliers.

- Min, Max, Range (x,y,z): These features capture the extremes and the range of variability in each axis and they are quite useful because the movements that we selected do vary in direction and intensity and with these 3 features, we could observe that jumping give a large Z range, whereas side-to-side movements have a high range on X axis whereas kicks tend to give one-sided spikes.

- Peak Frequency (x,y,z): This feature was obtained via FFT and is intended to capture the dominant frequency in the movement (amplitude of strongest frequency). This is good to know since walking for example has a regular rhythm whereas jumping has a higher z frequency.

- Spectral Entropy (x,y,z): The spectral entropy in each axis captures disorder or unpredictability in the frequency domain, one can think of it as how spread out the energy is across all frequencies obtained. In our

case, Rest has a low entropy whereas jumping and maybe walking tend to me more irregular and have a higher entropy.

- Mean Frequency (x,y,z): This feature was also obtained through the FFT and it captures the weighted average of the frequency content. Squats are slower (less acceleration too) and they tend to have lower mean frequency unlike jumping or kicking so this helped the model distinguish between fast explosive and slow rhythmic motions.
- FFT Energy (x,y,z): This feature refers to the total energy in the frequency spectrum and naturally activities like jumping have high energy, whereas rest has a quite low one.

Evaluation results for the tested classifiers and further optimization

10 machine learning algorithms were selected to test them for precision, recall and overall accuracy in order to obtain a decent classification system. This process was done with every single one of the three samples previously collected (one for every team member). The models tested are: Linear SVM, Radial Basis SVM, LDA, K-NN, Multilayer Perceptron (with over 2 layers), Gaussian Naive Bayes, Passive-Aggressive classifier, Bagging Classifier, Extra Trees Classifier, and Gaussian Process Classifier. From among the 10 algorithms, 2 of them stood out from the rest and yielded consistent results with all three samples.
The first one was the Multilayer Perceptron that yielded the following results for the best sample we obtained.

```
Results for MLP!

              precision    recall  f1-score   support

         1.0       0.98      1.00      0.99        60
         2.0       0.91      0.80      0.85        60
         3.0       0.95      0.93      0.94        60
         4.0       0.86      0.98      0.91        60
         5.0       0.80      0.80      0.80        60
         6.0       0.79      0.77      0.78        60

    accuracy                           0.88       360
   macro avg       0.88      0.88      0.88       360
weighted avg       0.88      0.88      0.88       360

Overall accuracy for MLP: 0.881
```

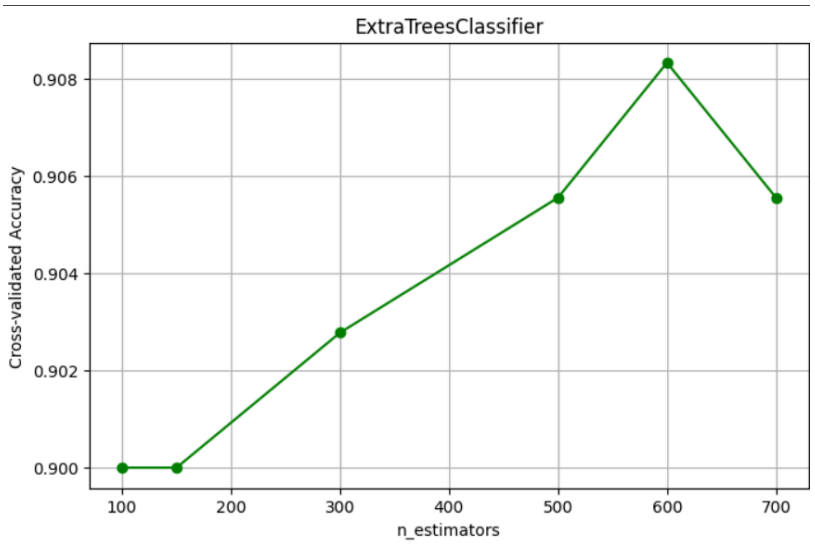
As we can tell from the report, it got precision and recall over .7 for all classes with most of them actually being in the high .8. Likewise, its overall accuracy was fairly decent at .881.

The multilayer perceptron model was only surpassed in results by the Extra Trees Classifier which got the following results for the exact same dataset:

```
Results for Extra Trees Classifier!

              precision    recall  f1-score   support

         1.0       1.00      1.00      1.00        60
         2.0       0.93      0.83      0.88        60
         3.0       0.98      0.98      0.98        60
         4.0       0.92      1.00      0.96        60
         5.0       0.85      0.73      0.79        60
         6.0       0.74      0.85      0.79        60

    accuracy                           0.90       360
   macro avg       0.90      0.90      0.90       360
weighted avg       0.90      0.90      0.90       360

Overall accuracy for Extra Trees Classifier: 0.900
```

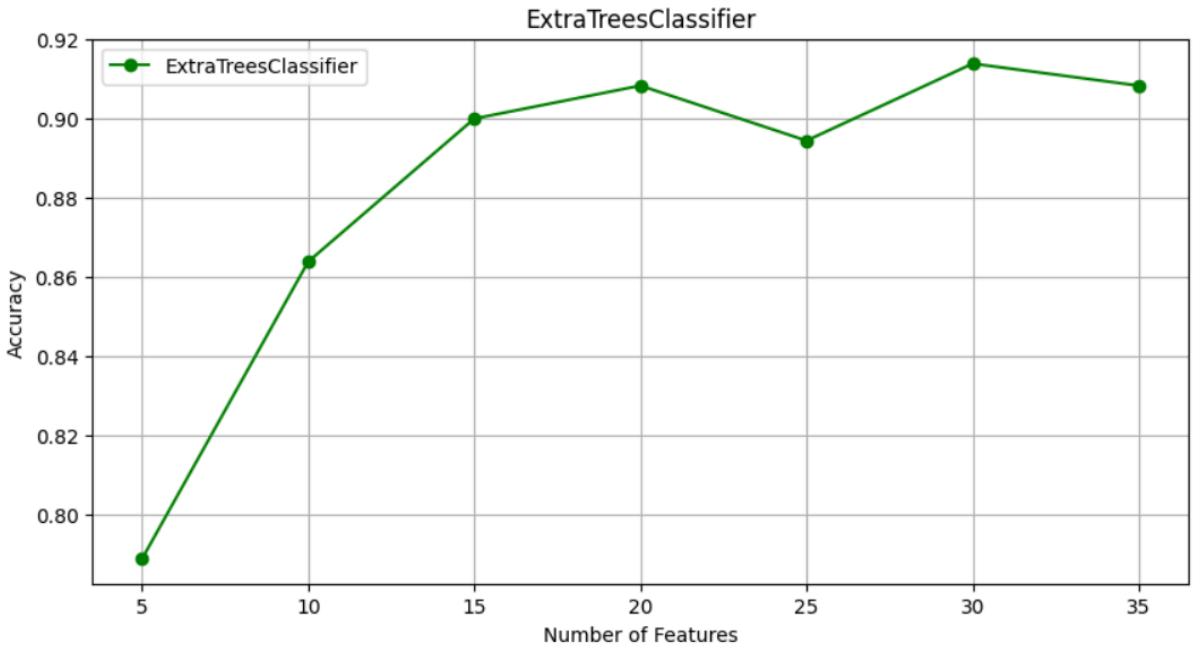
We can see that although it performed a little bit differently for some classes (5 and 6 mainly). Its overall accuracy was .9 which is slightly higher than that of the MLP and we can see some mild improvements in the first 3-4 classes.
Regarding all the other models tested, most of them had overall accuracies between .7 and .85 with some of them going as low as .6 for one of the team member's dataset. But, as mentioned previously, the superiority of MLP and ExtraTrees as well as the superiority of the latter over the former was consistent across the whole data bank.

After having selected the two best-performing models, we carried out a hyperparameter tuning and feature selection for both of them and then used nested cross validation to test the new models.
For our best-performing model (ExtraTreesClassifier), we tuned the parameter called "n-estimators" which refers to the number of decision trees created and we obtained the following results:



As we can see accuracy peaks at 600 trees.

Afterward, the feature selection process yielded the following results and in turn the features considered were: ['mean_x', 'std_x', 'mean_y', 'std_y', 'mean_z', 'std_z', 'rms', 'median_x', 'min_x', 'max_x', 'range_x', 'median_y', 'min_y', 'max_y', 'range_y', 'median_z', 'min_z', 'max_z', 'range_z', 'spectral_entropy_x', 'mean_freq_x', 'spectral_entropy_y', 'mean_freq_y', 'spectral_entropy_z', 'mean_freq_z']



Online classification results

For the online application, we downloaded the best performing model from the previous stages (ExtraTreesClassifier) and adapted the code provided by the professor to include our features and modifications so it was able to predict our activities accurately. The results we obtained from the online application were the ones we expected since our application was fully capable of detecting all 6 movements in real time with an overall accuracy that matched that of the cross-validation process previously obtained.

```
feature_names = [
    'mean_x', 'std_x', 'kurtosis_x', 'skew_x',
    'mean_y', 'std_y', 'kurtosis_y', 'skew_y',
    'mean_z', 'std_z', 'kurtosis_z', 'skew_z',
    'rms',
    'median_x', 'min_x', 'max_x', 'range_x',
    'median_y', 'min_y', 'max_y', 'range_y',
    'median_z', 'min_z', 'max_z', 'range_z',
    'peak_freq_x', 'spectral_entropy_x', 'mean_freq_x', 'fft_energy_x',
    'peak_freq_y', 'spectral_entropy_y', 'mean_freq_y', 'fft_energy_y',
    'peak_freq_z', 'spectral_entropy_z', 'mean_freq_z', 'fft_energy_z'
]

features_df = pd.DataFrame([features], columns=feature_names)

prediction = model.predict(features_df)

if prediction[0] == 1:
    print("Activity: Resting")
elif prediction[0] == 2:
    print("Activity: Jumping")
elif prediction[0] == 3:
    print("Activity: Squatting")
elif prediction[0] == 4:
    print("Activity: Side to side")
elif prediction[0] == 5:
    print("Activity: Kicking")
elif prediction[0] == 6:
    print("Activity: Walking")
else:
    print("")
```

Since we trained one model for each person, certain details varied but generally the models worked in all of our 3 cellphones, although the accuracy of the predictions depended on other factors slightly such as how tight or loose the pocket was in contrast to when the model was trained (sensitivity to acceleration) or even the position of the pocket in relation to the thigh because this changed the axes slightly. That being said, however, the model performed adequately in all cellphones and for every team member and the fact that the ExtraTreesClassifier had an accuracy of .9 was highly noticeable when it came to doing the same movements in real time. Further improvements could include adding even more sensors so it becomes capable of classifying more nuanced movements.

Reflections

Alan: This project was undoubtedly enlightening and fascinating in equal parts since we went ahead and applied all of the knowledge we acquired throughout the course regarding supervised learning models to a real-world scenario that could actually use the model we trained in order to make real-time predictions with a high accuracy. I personally found it quite rewarding to see how actual real data collected by ourselves performed differently with different models and then how we could squeeze their performance through optimization to finally get a functioning online app that works quite well. All in all, I feel satisfied with my participation as the person in charge of the model evaluation because for one, I learned a lot through this hands-on experience since I needed to understand the whole structure of the project. I can happily conclude that I have now been witness to the power that these models provide and their huge potential in real-life cases.

Pedro: Personally I really liked this project since it made us use all the topics we learned through the course and I liked that, since it was a way for us to wrap up everything we supposedly already knew and in case we forgot it made us relearn it, so I think that is really helpful. I found it fascinating and fun to do the data collection, we had to do it twice since the first time we thought that the reason the model wasn't performing as expected was because of a bad training (it ended up being something completely different) and because of this issue we had, it was really rewarding to see the model actually working and predicting correctly, kind of scary at first because of how accurate it was. Overall, I really liked this project and really enjoyed working as a part of the team. I think this project made me understand the course even more and I see it as a nice closure of it.

Miguel: I really enjoyed this project. It helped me greatly understand how supervised learning works, from data collection and model training to its real-time implementation. What surprised me most was that the models trained with our own data could actually recognize every movement we made. It was also surprising to learn how small changes, like the position of the phone, could have a huge impact on the model's performance. This made me realize how careful we must be when collecting data. I think the optimization part was one of the most complicated, since we had to test various hyperparameters and features until we found the best results, and this was extremely time-consuming. Finally, I am very happy with what I achieved

during this project, and it was a great opportunity to not only reinforce what we learned in class, but also to think outside the box to understand how to implement the models.

References

- *What is Samsung Health? | Samsung Health Tips | Samsung UK*. (2024, November 25). Samsung Uk. https://www.samsung.com/uk/mobile-phone-buying-guide/samsung-health-tips/

- *Google Fit*. (n.d.). Google Fit. https://www.google.com/fit/

- *ExtraTreesClassifier*. (n.d.). Scikit-learn. https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.ExtraTreesClassifier.html

- *A comparison of feature extraction methods for the classification of dynamic activities from accelerometer data*. (2009, March 1). IEEE Journals & Magazine | IEEE Xplore. https://ieeexplore.ieee.org/document/4663615

- Jia, N. (2025, January 21). *Spectral entropy – an underestimated time series feature*. Towards Data Science. https://towardsdatascience.com/spectral-entropy-an-underestimated-time-series-feature-94e18ae5b958/#:~:text=Fast%20Fourier%20Transform%20(FFT)%20is,value%20based%20on%20Shannon%20entropy.

- Wikipedia contributors. (2025, April 9). *Root mean square*. Wikipedia. https://en.wikipedia.org/wiki/Root_mean_square